National Textile University

Department of Computer Science

Subject:

Mobile application and development

Submitted To:

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Submitted By:

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Registration No:

23-NTU-CS-1221

Lab no:

4

Semester:

6th

Phase 1: The Static Structure

Task 1:

Task 1: The Safe Scaffold

- Create a StatelessWidget and return a Scaffold.
- Wrap the body in a SafeArea to prevent the UI from hitting the status bar or notch.

```dart
import 'package:flutter/material.dart';

void main() {
  runApp( MaterialApp(
    home:MyWidget()

  )

  );
}
class MyWidget extends StatelessWidget {
  const MyWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body:SafeArea(      //Task1

      )
    );


  }
}
```

Task 2:

- **Task 2: The Header (Circular Avatar & Margin)**
  - Use a Container with a margin (e.g., EdgeInsets.all(20.0)) to provide breathing room.
  - Place a CircleAvatar inside to represent the user's profile image.

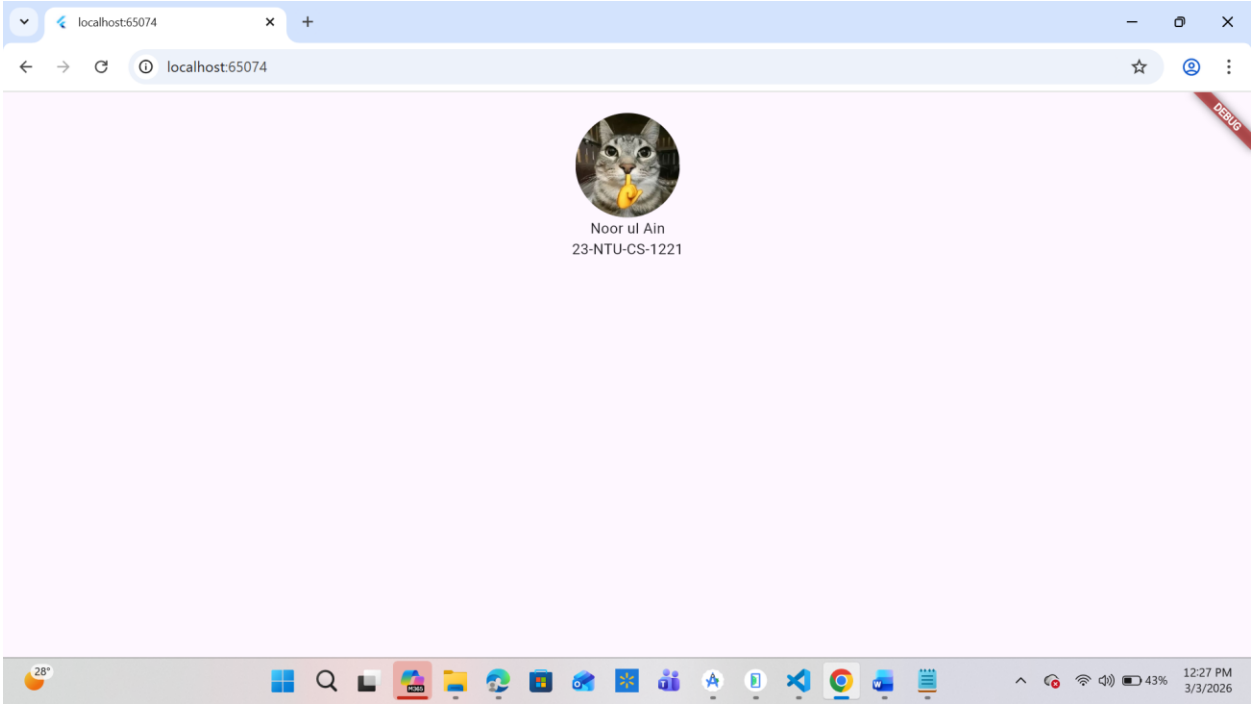
```dart
import 'package:flutter/material.dart';

void main() {
  runApp( MaterialApp(
    home:MyWidget()

  )

  );
}
class MyWidget extends StatelessWidget {
  const MyWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body:SafeArea(      //Task1
        child: Container(

          margin:EdgeInsets.all(20.0),
          child:CircleAvatar(
            radius:50,
            backgroundImage: AssetImage('assets/cat.jpg'),
          )
            )
        )
      );
  }
}
```

Task 3:

- **Task 3: Layout Alignment**
  - Use a Column to stack the Avatar and two Text widgets (Name and Title).
  - Set crossAxisAlignment to CrossAxisAlignment.center to align all elements to the middle.

```dart
import 'package:flutter/material.dart';

void main() {
  runApp( MaterialApp(
    home:MyWidget()

  )

  );
}
class MyWidget extends StatelessWidget {
  const MyWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body:SafeArea(      //Task1
        child: Container(         //Task 2 and 3

          margin:EdgeInsets.all(20.0),
          child:Column(
            crossAxisAlignment: CrossAxisAlignment.center,
            children :[CircleAvatar(
              backgroundImage: AssetImage('assets/cat.jpg')),
              Text('Noor ul Ain'),
              Text('23-NTU-CS-1221'),
            ]
          )
        )
      )

  );
  }
}
```

Output:



PHASE 2: PROPORTIONAL & FLEXIBLE LAYOUTS:

Task 4:

Task 4: The Body Content (Expanded & Proportions)

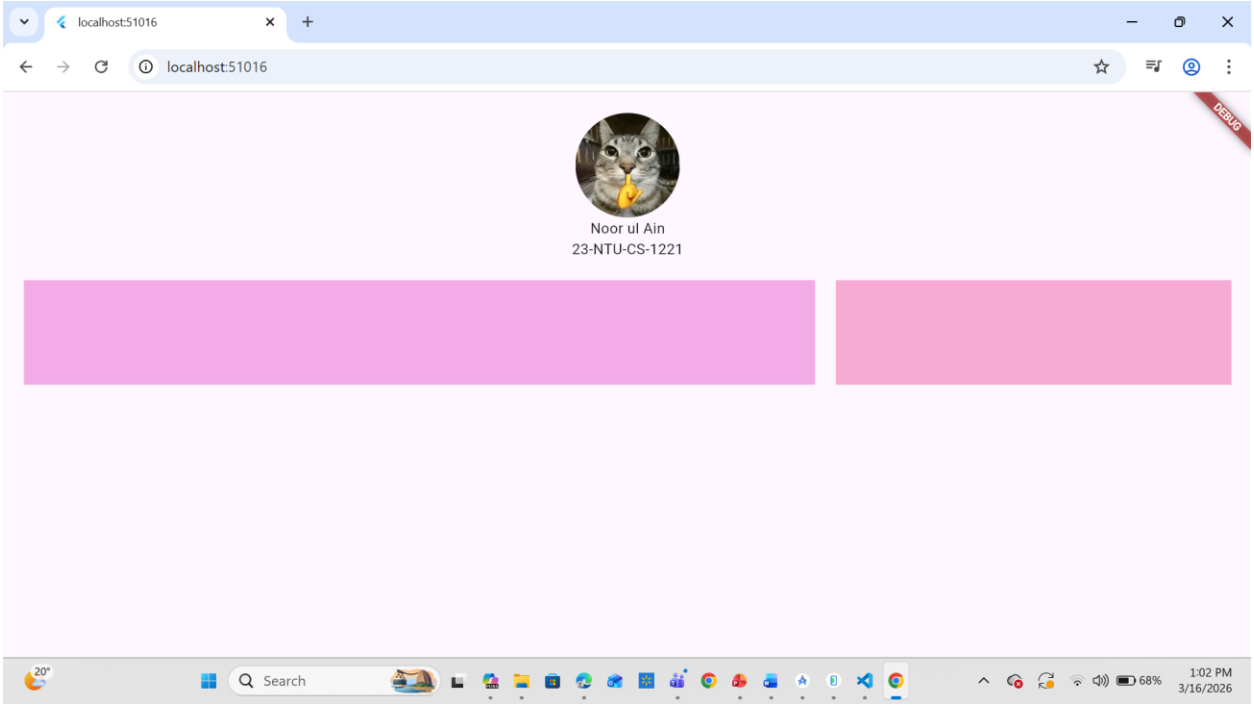
- Add a Row below the header inside the main Column.
- Insert two Container widgets inside this Row.
- Wrap both in Expanded widgets. Set the first container's flex to 2 and the second to 1 to create a 2:1 width ratio.

```
child:Column(
            crossAxisAlignment: CrossAxisAlignment.center,
            children :[CircleAvatar(
              radius:50,
              backgroundImage: AssetImage('assets/cat.jpg')),
              Text('Noor ul Ain'),
              Text('23-NTU-CS-1221'),
              SizedBox(
                  width:10,
                  height:10,
              ),
              //phase 2
              Row(           //Task4
                children:[
                Expanded(
                  flex:2,
                  child:Container(
                    height:100,
                    color: const Color.fromARGB(255, 243, 171, 231),
                  ),
                ),


                Expanded(
                  flex:1,
                  child:Container(
                    height:100,
                    color:Color.fromARGB(255, 245, 171, 210)
                  ),
                )
```

Task 5:

```
Row(            //Task4
                children:[
                Expanded(
                  flex:2,
                  child:Container(
                    height:100,
                    color: const Color.fromARGB(255, 243, 171, 231),
                  ),
                ),
                SizedBox(
                  height:10,
                  width:20,
                ),

                Expanded(
                  flex:1,

                  child:Container(
                    height:100,
                    color:Color.fromARGB(255, 245, 171, 210)
                  ),
                )
```

Output



PHASE 3: INTERACTIVITY & STATE MANAGEMENT

- **Task 6: Refactoring to StatefulWidget**
  - Convert the widget from Stateless to Stateful using the stf shortcut.
  - Define a boolean variable isFollowed and an integer score in the _State class.

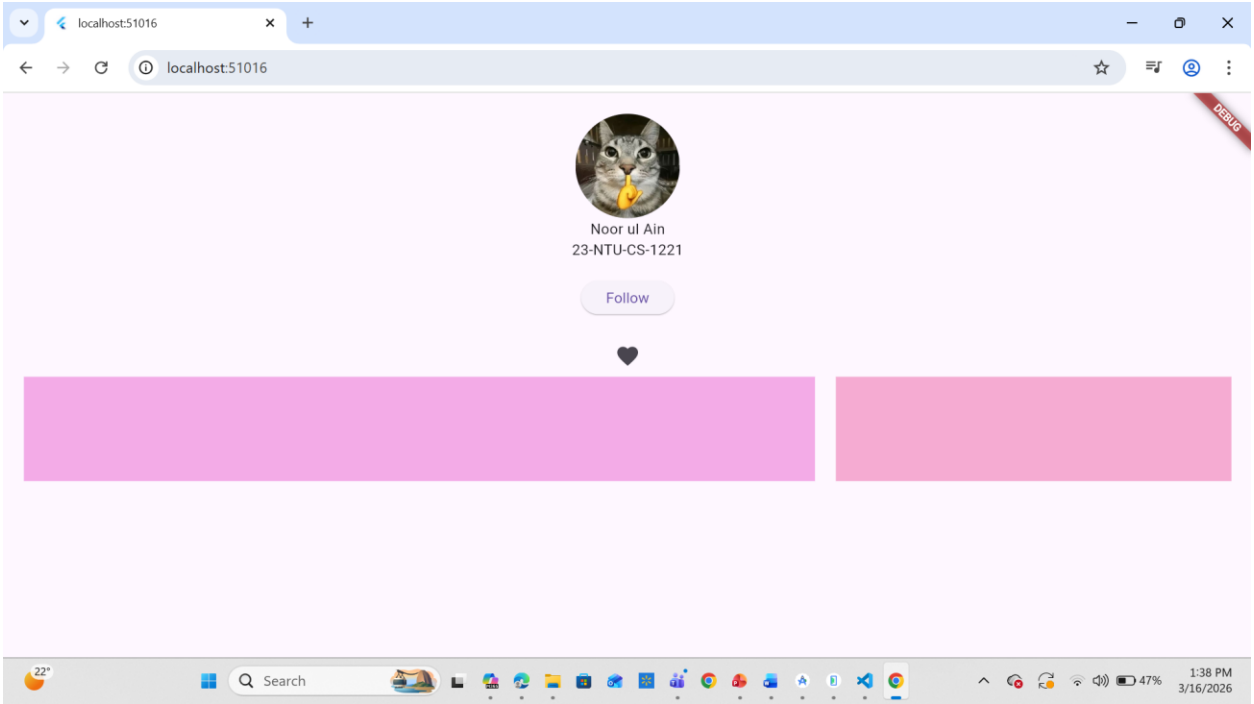
```dart
import 'package:flutter/material.dart';

void main() {
  runApp( MaterialApp(
    home:MyWidget()

  )

  );
}



class MyWidget extends StatefulWidget {        //Task: 6
  const MyWidget({super.key});
  @override
  State<MyWidget> createState()=> _MyWidgetState();
}
class _MyWidgetState extends State<MyWidget> {

  bool isFollowed = false;
  int score = 0;
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body:SafeArea(

        child: Container(

          margin:EdgeInsets.all(20.0),
         child:Center(
          child:Column(
            crossAxisAlignment: CrossAxisAlignment.center,
            children :[CircleAvatar(
              radius:50,
              backgroundImage: AssetImage('assets/cat.jpg')),
              Text('Noor ul Ain'),
              Text('23-NTU-CS-1221'),
              SizedBox(
                  width:20,
                  height:20,
              ),
              //phase 2
              Row(           //Task4
                children:[
                Expanded(
                  flex:2,
                  child:Container(
                    height:100,
                    color: const Color.fromARGB(255, 243, 171, 231),
                  ),
                ),
                SizedBox(
                  height:20,
                  width:20,
                ),

                Expanded(
                  flex:1,

                  child:Container(
                    height:100,
                    color:Color.fromARGB(255, 245, 171, 210)
                  ),
                )
                ]
              )
            ]
          )
        )
        )
      )


  );
  }
}
```

Task 7: Material Buttons & setState

- Add an ElevatedButton for "Follow" and an IconButton for "Like".
- In the onPressed callback, use setState() to toggle the isFollowed status or increment the score.

```
//Task 7:
                Row(
                 mainAxisAlignment: MainAxisAlignment.center,
                 children:[
                  ElevatedButton(
                    onPressed:(){
                    setState(() {
                        isFollowed = !isFollowed;
                    });},
                  child: Text(isFollowed ?"Following":"Follow"),
                  ) ]),
                SizedBox(
                    height:20,
                    width:20,
                   ),
                  IconButton(
                    icon:Icon(Icons.favorite),onPressed: (){
                      setState((){
                        score++;
                      }
                      );
                    },
                  ),
```

Output

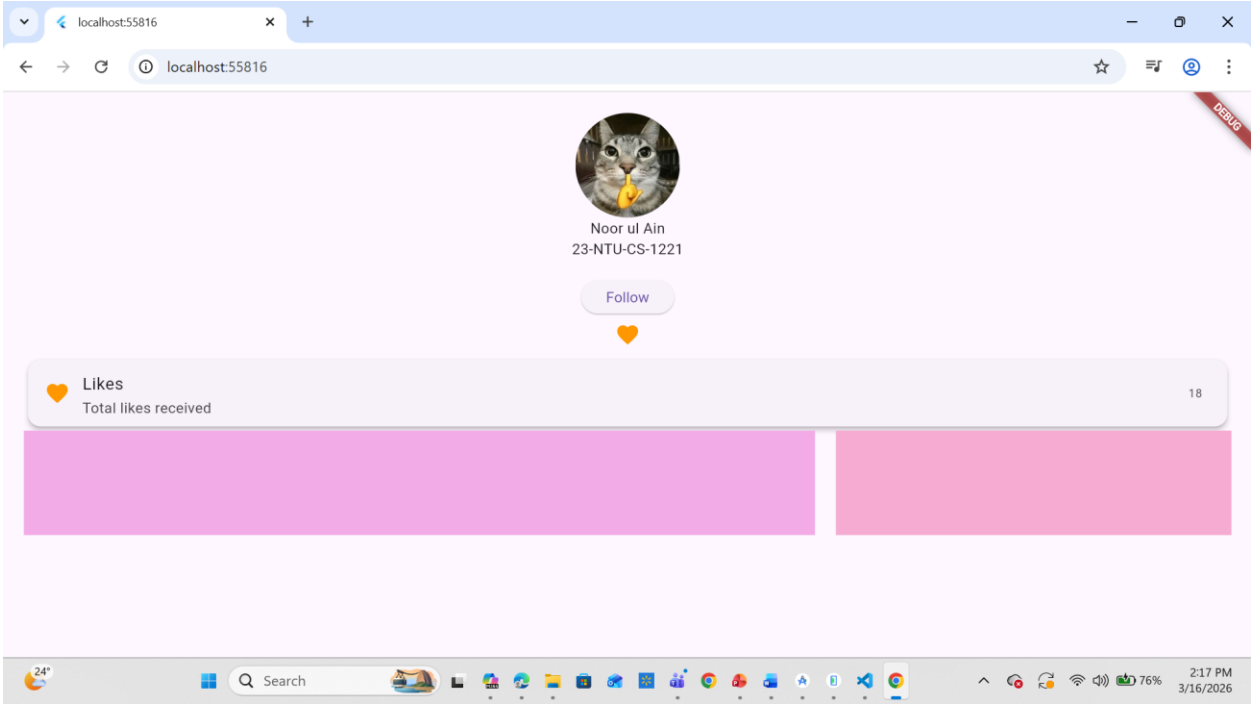


Task 8: Material Card Implementation

- Wrap the statistics (score) in a Card widget to give it a shadowed, elevated look.
- Use a ListTile inside the card for a clean "Icon + Title + Subtitle" arrangement.

```
//Task :8
              Card(
                elevation:4,
                child:ListTile(
                  leading:Icon(Icons.favorite,color:Colors.orange),
                  title: Text("Likes"),
                  subtitle: Text("Total likes received"),
                  trailing: Text('$score'),
                )
              ),
```

Output:


localhost:55816
localhost:55816
DEBUG
Noor ul Ain
23-NTU-CS-1221
Follow
Likes
Total likes received
18
Search
76%
2:17 PM
3/16/2026